CULLEN ABELSON

Pittsburgh, PA | cullen.abelson@pitt.edu | PERSONAL WEBSITE HERE

SCIENTIFIC INTERESTS

- **Delay time distributions** of stellar remnants (**CC** & **Ia SNe**, novae), as well as **WR stars** and red supergiants, to probe **binary stellar evolution**
- Galactic SED fitting using **prospector** to recover star formation history maps

EMPLOYMENT & EDUCATION

2022-PRESENT
PH.D. CANDIDATE
University of Pittsburgh | Department of Physics & Astronomy

2021-2022
POSTBACC. RESEARCH ASSISTANT
Penn State University | Department of Astronomy & Astrophysics

2017-2021
B.S. IN PHYSICS, B.S. IN ASTRONOMY
Penn State University | Schreyer Honors College

PUBLICATIONS | ORCID ID 0009-0007-2747-3691

05/2025
THE PROGENITOR SYSTEMS OF CLASSICAL NOVAE IN M31

C. S. Abelson; Carles Badenes; Laura Chomiuk; Benjamin F. Williams; Katelyn Breivik; L. Galbany; C. Jiménez-Palau
The Astrophysical Journal
DOI: 10.3847/1538-4357/adc68c

12/2024
DIAMANTE TESS AUTOREGRESSIVE PLANET SEARCH (DTARPS). III. UNDERSTANDING THE DTARPS-S CANDIDATE TRANSITING PLANET CATALOGS

Elizabeth J. Melton; Eric D. Feigelson; Marco Montalto; Gabriel A. Caceres; Andrew W. Rosenswie; Cullen S. Abelson
The Astronomical Journal
DOI: 10.3847/1538-3881/ad8355

08/2024
VIZIER ONLINE DATA CATALOG: DTARPS. II. 462 NEW TESS CANDIDATES

Melton, E. J.; Feigelson, E. D.; Montalto, M.; Caceres, G. A.; Rosenswie, A. W.; Abelson, C. S.
VizieR On-line Data Catalog
Bibcode: 2024yCat..51670203M

05/2024

DIAMANTE TESS AUTOREGRESSIVE PLANET SEARCH (DTARPS). I. ANALYSIS OF 0.9 MILLION LIGHT CURVES

Melton, Elizabeth J.; Feigelson, Eric D.; Montalto, Marco; Caceres, Gabriel A.; Rosenswie, Andrew W.; **Abelson, Cullen S.**
The Astronomical Journal
DOI: 10.3847/1538-3881/ad29f0

05/2024

DIAMANTE TESS AUTOREGRESSIVE PLANET SEARCH (DTARPS). II. HUNDREDS OF NEW TESS CANDIDATE EXOPLANETS

Melton, Elizabeth J.; Feigelson, Eric D.; Montalto, Marco; Caceres, Gabriel A.; Rosenswie, Andrew W.; **Abelson, Cullen S.**
The Astronomical Journal
DOI: 10.3847/1538-3881/ad29f1

10/2021

THE Z 2 [O III] LUMINOSITY FUNCTION OF GRISM-SELECTED EMISSION-LINE GALAXIES

Bowman, William P.; Ciardullo, Robin; Zeimann, Gregory R.; Gronwall, Caryl; Jeong, Donghui; Nagaraj, Gautam; **Abelson, Cullen**; Weiss, Laurel H.; Molina, Mallory; Schneider, Donald P.
The Astrophysical Journal
DOI: 10.3847/1538-4357/ac1a0e

OUTREACH

- Teen Astro Café (grades 9-12) — 02/2026
- Youth Astro Club (grades 3-5) — 02/2026
- High school science fair project advisor — 02/26 – present
- AstroPGH Python Boot Camp presenter — 05/2024, 05/2025
- Ask an Astronomer @ Allegheny Observatory Open House — 09/2025
- Astrophysicist for a Day @ Allegheny Observatory — 03/2024

AWARDS

- Dietrich School of Arts and Sciences Predoctoral Summer Research Fellowship — 05/2023 – 08/2023
- Thomas Lain Essay Competition Scholarship — 03/2023
- Penn State Academic Excellence Scholarship — 2017 – 2021
- Elsbach Honors Scholarship in Physics — 2019 – 2020
- Eagle Scout — 07/2017

TEACHING

- Private tutor: Introductory & intermediate physics — 10/2023 – present
- Teaching assistant: Introductory astronomy for non-majors — 08/2022 – 05/2023